

Distribution-Guided Policy Optimization for Fine-Grained Credit Assignment

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Abstract

Reinforcement learning is crucial for aligning large language models (LLMs) to perform complex reasoning tasks. However, current algorithms such as Group Relative Policy Optimization (GRPO) suffer from coarse-grained, sequence-level credit assignment, which severely struggles to isolate pivotal reasoning steps within long Chain-of-Thought generations. Furthermore, the standard unbounded Kullback-Leibler (KL) divergence penalty induces severe gradient instability and mode-seeking conservatism, ultimately stifling the discovery of novel reasoning trajectories. To overcome these limitations, we introduce Distribution-Guided Policy Optimization (DGPO), a novel critic-free reinforcement learning framework that reinterprets distribution deviation as a guiding signal rather than a rigid penalty. DGPO replaces the volatile KL divergence with the bounded Hellinger distance (an α -divergence) to safely quantify token-level exploration without the risk of gradient explosion. To effectively distinguish genuine reasoning breakthroughs from hallucinatory noise, we propose an entropy gating mechanism that scales this deviation by the policy’s epistemic uncertainty. By dynamically redistributing the coarse sequence-level advantage to individual tokens based on these gated scores, DGPO heavily incentivizes critical exploratory steps while suppressing unwarranted, low-entropy deviations. Consequently, DGPO completely eliminates the traditional token-level KL penalty and achieves fine-grained credit reallocation without the computational overhead of an additional value network. Extensive empirical evaluations demonstrate that DGPO sets a new state-of-the-art for critic-free alignment. Notably, on the Qwen2.5-32B architecture, DGPO achieves 60.0% Avg@32 accuracy and 46.0% Avg@32 accuracy on the challenging AIME2024 and AIME2025 benchmarks respectively, substantially outperforming competitive baselines like DAPO. Ultimately, our method provides a highly stable and theoretically grounded approach to unlock the deep reasoning potential of LLMs.

1 Introduction

The alignment of large language models (LLMs) through reinforcement learning (RL) has fundamentally advanced their capacity for complex problem-solving and logical reasoning [17, 7]. Recent paradigms, most notably Group Relative Policy Optimization [22] (GRPO), have demonstrated remarkable success in eliciting long-horizon Chain-of-Thought (CoT) behaviors without relying on an auxiliary value network. By optimizing a policy relative to a group of sampled responses, GRPO drastically reduces memory overhead while maintaining competitive performance on mathematical and algorithmic reasoning tasks [7]. In this setting, models learn to self-correct and explore extended logical trajectories purely through scalar rewards provided by rule-based verifiers or reward models.

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Despite these architectural advantages, current critic-free RL frameworks suffer from a severe coarse credit assignment problem [26, 20]. In standard GRPO, a single scalar advantage is uniformly broadcasted across the entire generated sequence, which often spans thousands of tokens. This coarse-grained sequence-level assignment forces the model to treat pivotal reasoning breakthroughs and redundant transitional syntax with equal importance, highly diluting the learning signal. Compounding this inefficiency is the standard reliance on the Kullback-Leibler [11, 39] (KL) divergence penalty to prevent the policy from collapsing. Specifically, the unbounded Reverse KL divergence enforces extreme mode-seeking conservatism; it indiscriminately and harshly penalizes any deviation from the reference policy. When the model attempts to explore novel, low-probability reasoning trajectories, this unbounded penalty causes severe gradient instability. Consequently, the model is disincentivized from discovering innovative solutions, trapping its reasoning capabilities within the safe, yet restricted, bounds of the pre-trained distribution [10].

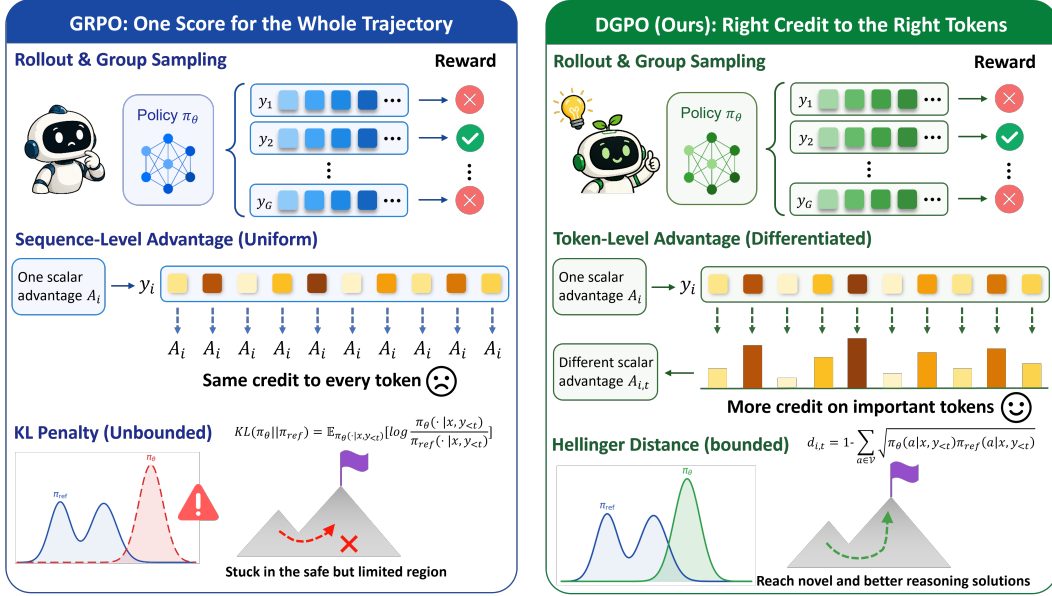


Figure 1: Conceptual comparison between standard GRPO and our proposed DGPO. While GRPO uniformly broadcasts a coarse-grained sequence-level advantage and imposes an unbounded Reverse KL penalty that stifles exploration, DGPO dynamically reallocates advantages to individual tokens.

To overcome these theoretical and practical bottlenecks, we propose Distribution-Guided Policy Optimization (DGPO), a novel RL framework that reallocates coarse sequence-level advantages into fine-grained, token-level update signals. DGPO operates on a novel perspective: deviation from the reference distribution should not be uniformly penalized as an error, but rather interpreted as a guiding signal for exploration. When a sequence yields a positive reward, the tokens that diverge most from the reference model are often the exact cognitive leaps responsible for the success. To safely quantify this deviation, DGPO replaces reverse KL divergence with the bounded Hellinger distance [2, 1, 28]. Unlike Reverse KL, this bounded metric smoothly encourages exploratory diversity without the risk of gradient explosion during training.

However, relying solely on distribution deviation risks rewarding "fake innovations." In particular, a large deviation from the reference policy does not necessarily indicate a meaningful reasoning step—it may also arise from spurious or hallucinated tokens. For example, when the model assigns high confidence to an incorrect, out-of-distribution token, the resulting deviation can be large despite lacking semantic validity. Such confident hallucinations are particularly problematic, as they would be incorrectly reinforced by deviation-based signals alone. To mitigate this, DGPO introduces a novel policy entropy gating mechanism. Scaling the Hellinger distance by the model’s normalized epistemic uncertainty (entropy), the algorithm successfully distinguishes deliberate, high-uncertainty exploration from confident, low-entropy hallucinations. These gated scores are then used to dynamically redistribute the sequence-level advantage across individual tokens. Critical exploratory steps receive amplified gradients, while standard syntactic tokens are naturally discounted.

To empirically validate our framework, we conduct extensive evaluations on the highly challenging AIME 2024 and AIME 2025 mathematical reasoning benchmarks. Evaluated on model architectures of various sizes, DGPO establishes a new state-of-the-art for critic-free alignment methods. Specifically, DGPO achieves an impressive 60.0% Avg@32 accuracy on AIME 2024 and 46.0% on AIME 2025, substantially outperforming the highly competitive DAPO baseline. Beyond absolute outcome performance, our computational profiling demonstrates that DGPO preserves the uncompromising hardware advantages of standard GRPO. By completely bypassing the need for an auxiliary value network, DGPO achieves process-level supervisory signals with a marginal time overhead of only 3.6% and a memory footprint strictly comparable to vanilla GRPO.

In summary, our main contributions are three-fold: (i) We mathematically identify the limitations of uniform sequence-level updates and the unbounded Reverse Kullback-Leibler (KL) divergence in long-horizon tasks, revealing how they induce mode-seeking conservatism and severe gradient instability. (ii) We propose a novel, critic-free RL framework that reinterprets distribution deviation as a guiding signal. By synergizing the bounded Hellinger distance with a policy entropy gating mechanism, DGPO safely incentivizes critical exploratory steps while filtering out confident hallucinations. (iii) We demonstrate that DGPO substantially elevates the reasoning capabilities of LLMs across different model scales (7B and 32B) on demanding mathematical benchmarks. It achieves fine-grained credit reallocation and faster convergence while maintaining the highly scalable computational efficiency of sequence-level objectives.

2 Related Works

2.1 Reinforcement Learning for LLM Reasoning

The application of reinforcement learning (RL) has become the de facto standard for aligning large language models (LLMs) with human intent and enhancing their complex reasoning capabilities [7, 17]. While Proximal Policy Optimization (PPO) initially dominated the landscape of RL from Human Feedback (RLHF) [21], its reliance on an auxiliary value network imposes prohibitive memory and computational overheads, particularly as model parameters scale. This architectural bottleneck has driven a paradigm shift towards critic-free optimization methods [6, 7, 8, 9, 19, 33, 37]. Among these, Group Relative Policy Optimization (GRPO) [22] has demonstrated exceptional efficacy in eliciting advanced mathematical and logical reasoning, notably empowering models to perform extensive Chain-of-Thought (CoT) generation [38]. By estimating the baseline through group-wise reward normalization rather than a parameterized critic, GRPO significantly streamlines the training pipeline. However, GRPO applies a uniform, sequence-level advantage to all tokens within a response [12], treating every decoding step equally. In the context of long-horizon reasoning, this coarse-grained optimization inevitably obscures the learning signal and slows convergence.

2.2 Fine-Grained Credit Assignment

The credit assignment problem is a fundamental challenge in RL, uniquely exacerbated in autoregressive LLMs [20], where action spaces span tens of thousands of tokens. To provide more granular supervisory signals, recent literature has heavily investigated Process Reward Models (PRMs) [14, 27, 35, 23, 32, 15]. Unlike Outcome Reward Models (ORMs) [4, 3, 36, 5, 31] that evaluate only the final answer, PRMs assign explicit scalar rewards to intermediate reasoning steps. Despite their effectiveness, training robust PRMs requires exhaustive human-annotated step-level data. Alternatively, several works have attempted to redistribute sequence-level rewards using attention weights or heuristic decay mechanisms [34, 13]. Nevertheless, these approximations often lack rigorous theoretical justification and fail to isolate the true pivotal steps of a reasoning trajectory. Our proposed DGPO bypasses the need for external PRMs and parameterized critics entirely. By extracting inherent probabilistic deviations between the current and reference policies, DGPO establishes a theoretically grounded, self-contained mechanism for token-level credit assignment.

3 Distribution-Guided Policy Optimization

3.1 Preliminaries and Motivation

In the context of aligning large language models (LLMs) for complex reasoning tasks, such as Chain-of-Thought generation, standard Group Relative Policy Optimization (GRPO) relies on a coarse-grained, sequence-level reward mechanism. Given a prompt x , the policy π_θ generates a group of candidate responses $\{y_1, \dots, y_G\}$, which are subsequently evaluated to yield scalar rewards $\{r_1, \dots, r_G\}$. GRPO computes a sequence-level advantage:

$$A_i = (r_i - \hat{\mathbb{E}}[r]) / (\widehat{\text{Var}}[r]^{1/2} + \varepsilon) \quad (1)$$

through group-wise Z-score normalization. However, this paradigm suffers from two critical limitations. First, the advantage A_i is uniformly distributed across all tokens in a potentially long sequence, making it impossible for the model to distinguish pivotal reasoning breakthroughs from redundant transitional phrases. Second, the standard objective incorporates an unbounded Kullback-Leibler (KL) divergence penalty to constrain the policy to a reference model π_{ref} . When the policy explores highly novel trajectories where π_{ref} assigns near-zero probability, the KL penalty can explode, leading to severe gradient instability. To address these issues, we propose Distribution-Guided Policy Optimization (DGPO), which reinterprets distribution deviation not as a penalty, but as a guiding signal for critic-free, fine-grained credit assignment.

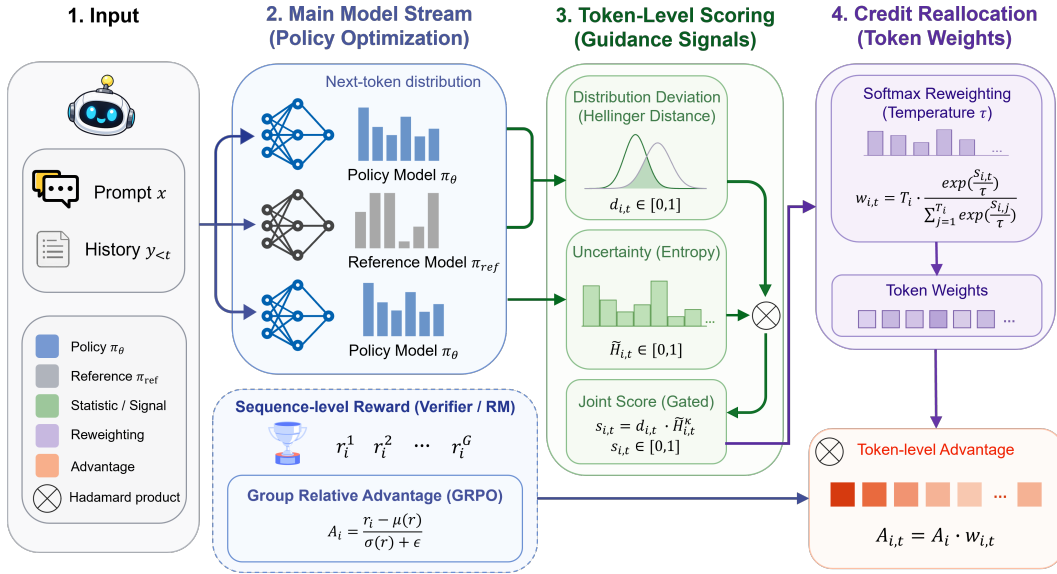


Figure 2: The computational pipeline of Distribution-Guided Policy Optimization (DGPO).

3.2 Distribution-Guided Advantage Redistribution

Instead of treating distribution constraints as an extrinsic penalty term (e.g., the KL-divergence in standard PPO or GRPO), DGPO internalizes the regularization by reinterpreting distribution deviation as a token-level guidance signal. This approach transforms the coarse sequence-level advantage A_i into a fine-grained credit assignment via a redistribution weight $w_{i,t}$.

A primary challenge in RLHF [17, 39] is the numerical instability caused by the unbounded nature of the Kullback-Leibler (KL) divergence, especially when the current policy π_θ explores regions where the reference policy π_{ref} assigns near-zero probability. To mitigate this, we employ the Hellinger distance, to quantify the deviation at each step t :

$$d_{i,t} = H^2(\pi_\theta(\cdot|x, y_{<t}), \pi_{ref}(\cdot|x, y_{<t})) = 1 - \sum_{a \in \mathcal{V}} \sqrt{\pi_\theta(a|x, y_{<t}) \pi_{ref}(a|x, y_{<t})} \quad (2)$$

where $d_{i,t} \in [0, 1]$. The intrinsic boundedness of the Hellinger distance ensures that the magnitude of the guidance signal remains saturated even during aggressive exploration, preventing the "penalty explosion" common in traditional KL-based objectives.

However, relying solely on distribution deviation makes the algorithm susceptible to "fake innovations," such as the model confidently hallucinating a rare, out-of-distribution word [29, 25]. To filter out such noise, DGPO introduces a policy entropy gating mechanism. The redistribution weight $w_{i,t}$ is defined as:

$$w_{i,t} = \text{Softmax}_t \left(\frac{d_{i,t} \cdot \mathcal{H}(\pi_\theta(\cdot|x, y_{<t}))}{\tau} \right) \cdot T_i \quad (3)$$

where $\mathcal{H}(\cdot)$ denotes the Shannon entropy and τ is a temperature hyperparameter. The inclusion of T_i (sequence length) ensures the unit-mean property $\frac{1}{T_i} \sum_t w_{i,t} = 1$, preserving the overall gradient magnitude of the group.

Gradient Stability Analysis. The technical superiority of DGPO lies in its gradient dynamics. In standard KL-penalized RL, the gradient of the objective:

$$\mathcal{L}_{KL} = \hat{\mathbb{E}}[\rho_t A_t - \beta \text{KL}(\pi_\theta || \pi_{ref})] \quad (4)$$

w.r.t. θ involves a term $\beta \nabla_\theta \log \pi_\theta(\log \frac{\pi_\theta}{\pi_{ref}} + 1)$. As $\pi_{ref} \rightarrow 0$, this term becomes unbounded, inducing severe gradient spikes. In contrast, the gradient of the DGPO objective (Eq 7) can be approximated as:

$$\nabla_\theta \mathcal{L}^{DGPO} \approx \mathbb{E}[w_{i,t} \cdot A_i \cdot \nabla_\theta \log \pi_\theta] \quad (5)$$

Here, the divergence signal $d_{i,t}$ enters the optimization solely as a multiplicative factor within $w_{i,t}$. Since $w_{i,t}$ is strictly bounded by construction (e.g., via the softmax normalization and the bounded Hellinger distance), the gradient norm $\|\nabla_\theta \mathcal{L}^{DGPO}\|$ is essentially governed by the group-relative advantage A_i and the score function. By remapping the "unbounded penalty" into a "bounded reweighting factor," DGPO achieves a stable optimization landscape without sacrificing the regularization effects necessary for policy alignment. The complete gradient analysis is shown in the appendix A.1.

3.3 Fine-Grained Credit Reallocation and Objective

The final step in DGPO is to reallocate the sequence-level advantage A_i to individual tokens based on their gated scores. To preserve the overall magnitude of the optimization signal—thereby preventing gradient vanishing across long sequences—we transform the scores $s_{i,t}$ into token-level importance weights $w_{i,t}$ using a temperature-scaled softmax explicitly scaled by the sequence length T_i . Formally, we define $w_{i,t}$ as:

$$w_{i,t} = T_i \cdot \frac{\exp(s_{i,t}/\tau)}{\sum_{j=1}^{T_i} \exp(s_{i,j}/\tau)} \quad (6)$$

which guarantees that the mean of the weights across the sequence is strictly 1. The coarse-grained advantage is then redistributed to yield the fine-grained local advantage $A_{i,t} = A_i \cdot w_{i,t}$. Through this reallocation, pivotal exploratory tokens receive amplified update signals, while standard grammatical tokens receive diminished gradients. Because the reference distribution is elegantly internalized within the localized advantage, DGPO bypasses the need for standard token-level KL penalty from the loss function. The final DGPO objective is thus formulated as:

$$L^{DGPO}(\theta) = \frac{1}{G} \sum_{i=1}^G \left(\frac{1}{T_i} \sum_{t=1}^{T_i} \min(\rho_{i,t} A_{i,t}, \text{clip}(\rho_{i,t}, 1 - \epsilon_c, 1 + \epsilon_c) A_{i,t}) \right) \quad (7)$$

where $\rho_{i,t}$ is the standard importance sampling ratio: $\rho_{i,t} = \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{old}}(y_{i,t}|x, y_{i,<t})}$, calculated between the current policy π_θ and the behavior policy $\pi_{\theta_{old}}$. By leveraging its own probabilistic dynamics, DGPO achieves stable, critic-free token-level credit assignment, fully unlocking the reasoning potential of LLMs in long-sequence generation.

4 Experiment

4.1 Experimental Setup

To ensure a rigorous and strictly controlled comparison, we evaluate Distribution-Guided Policy Optimization (DGPO) under the exact training settings established by recent large-scale open-source RL frameworks [24]. We adopt Qwen2.5-32B-Base [18] as our primary backbone, deliberately selecting a model with no prior exposure to long-form Chain-of-Thought (CoT) synthetic data. This guarantees that any emergence of deep reasoning stems strictly from our policy optimization algorithm rather than pre-distilled priors. Furthermore, we conduct preliminary scaling and ablation studies on Qwen2.5-7B-Math [30]. All models are trained exclusively on the publicly released DAPO-17K dataset [33].

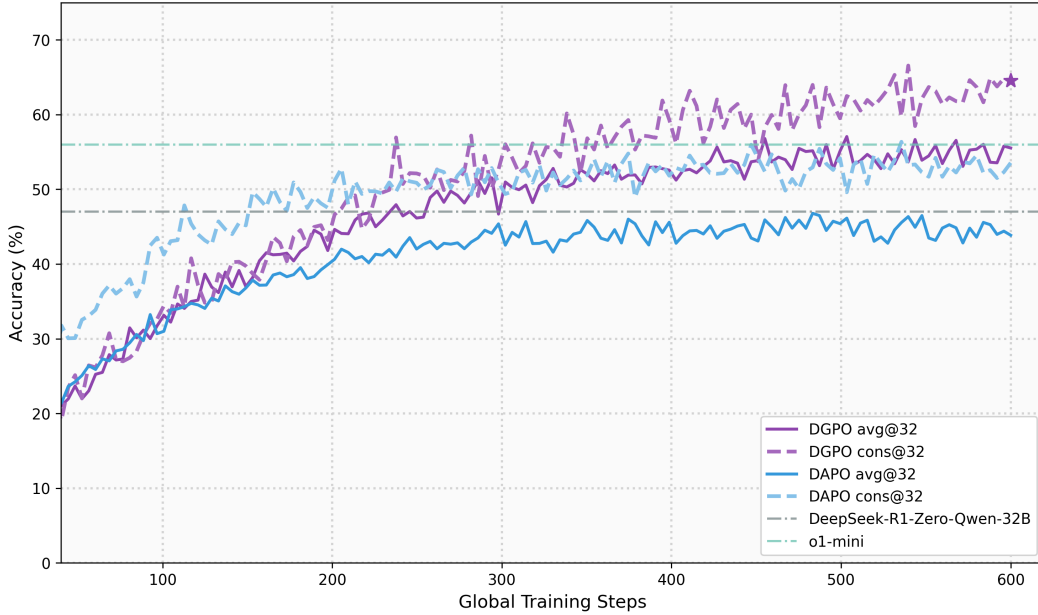


Figure 3: Validation accuracy on the AIME benchmark during training (Qwen2.5-32B-Base). The learning curves illustrate the progression of reasoning performance over global training steps. We report both the average Pass@1 and consensus accuracy.

During the optimization of the 32B model, we maintain a global batch size of 512 prompts, sampling a group size of $G = 16$ responses per prompt. The policy is optimized using a learning rate of 1×10^{-6} and a weight decay of 0.1. To ensure stable gradient estimation during the token-level advantage reallocation, we employ a mini-batch size of 64 prompts (1,024 samples) per update.

For evaluation, we utilize the highly challenging AIME 2024 benchmark as our primary validation suite, supplemented by AIME 2025 to test the absolute problem-solving boundaries. To account for the inherent variance in autoregressive CoT generation, we rigorously follow the evaluation protocol of repeating inference 32 times per problem. We report the average Pass@1 (Avg@32) as the most robust indicator of reasoning reliability, alongside the majority vote (Cons@32) and the probability of at least one correct answer (Pass@32). All inference evaluations are conducted with a temperature of 1.0 and a top-p threshold of 0.7.

4.2 Main Results

We benchmark DGPO against leading critic-free RL algorithms, including the standard GRPO formulation and the highly competitive DAPO baseline. Table 1 summarizes the reasoning performance on the AIME benchmarks. On the 32B scale, DGPO systematically breaks the performance ceiling of standard outcome-based reward methods, such as standard GRPO and DAPO. Evaluated on AIME 2024, our method achieves a robust average Pass@1 accuracy of 60.0%, substantially outperforming

the DAPO baseline which plateaus at 50.0%. This performance leap is consistently reflected in the consensus metric (Cons@32), which surges from 60.0% to 73.0%. On the significantly more demanding AIME 2025 benchmark, DGPO maintains its superiority, elevating the Avg@32 metric from 38.0% to 46.0%.

Method	AIME 2024			AIME 2025		
	Avg@32	Cons@32	Pass@32	Avg@32	Cons@32	Pass@32
DAPO [33]	50.0%	60.0%	80.0%	38.0%	47.0%	63.0%
FIPO [16]	56.0%	73.0%	83.0%	43.0%	50.0%	67.0%
DGPO (Ours)	60.0%	73.0%	87.0%	46.0%	53.0%	67.0%

Table 1: Comparison of reasoning performance on AIME benchmarks. All results are reported as percentages (%). To align with prior baseline reporting and reduce sensitivity to digit-level generation variance, final values are rounded to the nearest integer.

The efficacy of our token-level credit assignment is further corroborated by our experiments on the Qwen2.5-7B-Math model [30]. Despite the limited parameter scale and restricted intrinsic search space, DGPO achieves a notable Pass@1 accuracy of 43.0% on AIME 2024. This fundamentally eclipses the vanilla GRPO baseline, which struggles at 22.0%, and comfortably surpasses the 36.0% achieved by DAPO. These results confirm that by safely localizing advantages using entropy-gated α -divergence, DGPO can extract dramatically richer supervisory signals from the same binary outcome rewards, optimizing reasoning capabilities across different model scales.

Method	AIME 2024 (Pass@1)	AIME 2025 (Pass@1)
GRPO [22]	22.0%	18.0%
DAPO [33]	36.0%	18.0%
FIPO [16]	40.0%	19.0%
DGPO (Ours)	43.0%	24.0%

Table 2: Comparison of Pass@1 Performance on Qwen2.5-7B-MATH. All results are reported as percentages (%) representing the peak average Pass@1 across 32 samples (Avg@32).

4.3 Ablation Studies

To rigorously validate the contribution of each core component within the DGPO framework, we conduct extensive ablation studies on the Qwen2.5-7B-Math model. We systematically isolate the effects of the Hellinger distance substitution and the policy entropy gating mechanism.

Method	AIME 2024 (Pass@1)	AIME 2025 (Pass@1)
DGPO (Full)	43.0%	24.0%
DGPO (w/o Entropy Gate)	38.0%	20.0%
DGPO (Reverse KL instead of Hellinger)	34.0%	18.0%

Table 3: Ablation study of DGPO components on Qwen2.5-7B-MATH. All results are reported as percentages (%) representing the peak average Pass@1 across 32 samples (Avg@32).

Impact of the Entropy Gating Mechanism. DGPO introduces a policy entropy gating mechanism to filter out "fake innovations" and hallucinations. To evaluate its necessity, we remove the entropy scaling ($\kappa = 0$) and rely solely on the token-level distribution deviation. As shown in Table 3, the removal of the entropy gate leads to a performance degradation of 5% on the AIME 2024 benchmark. This empirically validates our hypothesis that unconstrained deviation often rewards overly confident hallucinations, and scaling by normalized epistemic uncertainty is critical for stable exploration.

Hellinger Distance vs. Unbounded KL. A fundamental claim of DGPO is that replacing the volatile Kullback-Leibler (KL) divergence with the bounded Hellinger distance prevents mode-seeking conservatism. We replace our Hellinger distance metric with the standard Reverse KL divergence

while keeping the token-level reallocation active. The results demonstrate that the Reverse KL variant suffers from early convergence to suboptimal local minima, yielding a Pass@1 of only 34.0%. This confirms that the bounded nature of the Hellinger distance safely encourages exploratory diversity.

4.4 Hyperparameter Sensitivity Analysis

Temperature (τ)	Pass@1 (%)
$\tau = 0.1$ (Over-concentrated)	35.0
$\tau = 0.5$	43.0
$\tau = 1.0$	42.0
$\tau = 5.0$ (Near-uniform)	37.0

Table 4: Sensitivity to Reallocation Temperature (τ). Evaluated on Qwen2.5-7B (AIME 2024).

Entropy Scale (κ)	Pass@1 (%)
$\kappa = 0.0$ (w/o Gating)	38.0
$\kappa = 0.5$	41.0
$\kappa = 1.0$	43.0
$\kappa = 2.0$	40.0
$\kappa = 5.0$ (Over-penalized)	36.0

Table 5: Sensitivity to Entropy Gating Factor (κ). Evaluated on Qwen2.5-7B (AIME 2024).

To comprehensively evaluate the robustness of DGPO, we analyze its sensitivity to two core hyperparameters: the reallocation temperature τ and the entropy gating scaling factor κ . All sensitivity experiments are conducted on the Qwen2.5-7B-Math backbone using the AIME 2024 benchmark.

Reallocation Temperature (τ). The temperature hyperparameter τ controls the sharpness of the fine-grained credit reallocation in the softmax transformation. We evaluated DGPO across varying temperature scales, shown in Tab 4. When $\tau \rightarrow \infty$, the token-level importance weights $w_{i,t}$ naturally regress towards a uniform distribution. This effectively degrades the localized token-level advantage $A_{i,t}$ back to coarse-grained sequence-level advantage A_i , diluting the supervisory signal. Conversely, an excessively low temperature (e.g., $\tau = 0.1$) induces highly sparse and abrupt update signals, leading to extreme gradient variance and optimization instability. DGPO achieves optimal and stable credit assignment within a moderate range (e.g., $\tau \in [0.5, 1.0]$), confirming that our framework safely incentivizes critical exploratory steps without requiring exhaustive, task-specific tuning.

Entropy Gating Scaling Factor (κ). The scaling factor κ governs the strictness of the policy entropy gating mechanism, defined as $s_{i,t} = d_{i,t} \cdot \tilde{H}_{i,t}^\kappa$. This mechanism is crucial for filtering out confident hallucinations by scaling the Hellinger Distance deviation. As demonstrated in Table 5, setting $\kappa = 0$ causes the algorithm to erroneously reward low-entropy, out-of-distribution tokens, degrading the Pass@1 accuracy to 38.0%. On the other end of the spectrum, setting κ to an excessively high value overly penalizes valid exploratory steps unless the model is in a state of maximum epistemic uncertainty. This excessively strict gating dampens the guidance signal provided by the distribution deviation. We find that a moderate value strikes the optimal balance: it aggressively suppresses spurious noise while preserving the gradient magnitude of genuine reasoning breakthroughs, ensuring stable alignment during long-horizon Chain-of-Thought (CoT) generation.

4.5 Computational Profiling

A major advantage of critic-free RL frameworks is their memory and computational efficiency. To rigorously demonstrate that DGPO preserves these hardware advantages while achieving fine-grained credit assignment, we profile the peak memory allocation and training throughput during the alignment of the Qwen2.5-7B model. As summarized in Table 6, we compare DGPO against standard GRPO and a traditional PPO implementation that requires an auxiliary parameterized value network.

Method	Peak Memory (GB)	Throughput (Tokens/s/GPU)	Time Overhead
PPO (w/ Critic)	72.4	85	+129.4%
GRPO (Baseline)	46.2	195	0.0%
DGPO (Ours)	46.5	188	+3.6%

Table 6: Computational profiling on Qwen2.5-7B during RL training. Metrics are measured on 2 nodes with $8 \times$ H20 (96GB) GPUs using DeepSpeed ZeRO-3.

The traditional PPO algorithm imposes a prohibitive memory bottleneck, consuming approximately 72.4 GB per GPU and severely bottlenecking throughput due to the forward and backward passes required for the critic model. In contrast, both GRPO and DGPO operate within a highly streamlined pipeline, fundamentally reducing the memory footprint to roughly 46 GB per GPU.

Crucially, the additional computational overhead introduced by DGPO is nearly negligible compared to the vanilla GRPO baseline. In standard GRPO, computing the sequence-level Kullback-Leibler (KL) divergence already requires obtaining the full vocabulary logits from both the current policy π_θ and the reference policy π_{ref} . DGPO completely eliminates this token-level KL penalty. Instead, it utilizes these exact same logit tensors to compute the Hellinger distance (Equation 6) and the normalized policy entropy. Because these operations—such as element-wise square roots and token-level softmax accumulations—are highly parallelizable tensor computations executed directly on the GPU, they bypass the need for any additional neural network forward passes.

Empirically, DGPO achieves a training throughput of 188 tokens/s/GPU, introducing a marginal time overhead of only 3.6% compared to the coarse-grained GRPO baseline. This profiling solidifies DGPO as a highly scalable solution: it successfully extracts the rich, fine-grained supervisory signals typically associated with Process Reward Models [23] or PPO [21], but executes with the uncompromising computational efficiency of a purely sequence-level, critic-free objective.

4.6 Qualitative Analysis: Token-Level Credit Assignment

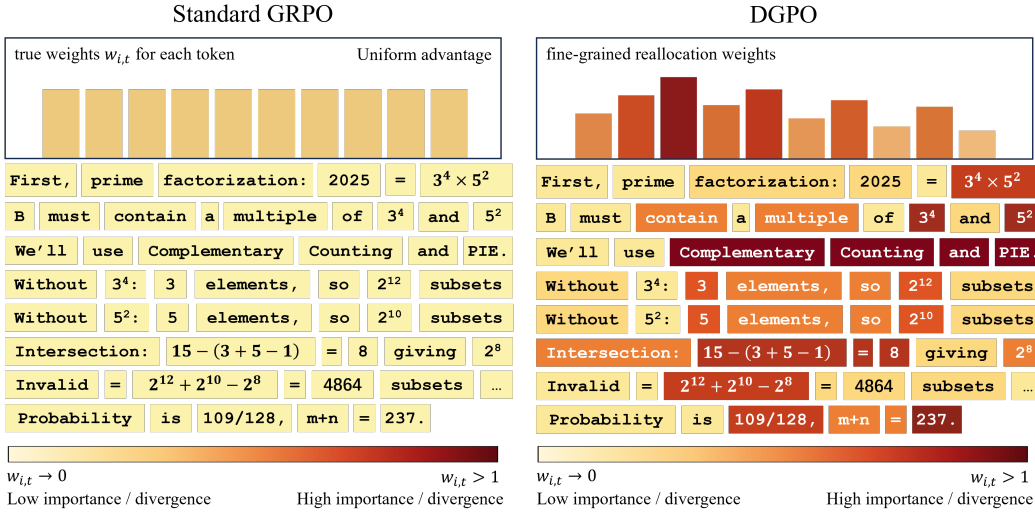


Figure 4: Qualitative visualization of the token-level credit reallocation. The background color intensity corresponds to the magnitude of the redistributed importance weight $w_{i,t}$.

To intuitively understand how DGPO achieves fine-grained credit reallocation, we visualize the redistributed token-level importance weights $w_{i,t}$ defined in Figure 4.

When analyzing successful long-horizon Chain-of-Thought (CoT) generations, we observe a stark contrast between standard sequence-level assignment and our method. In standard GRPO, a single scalar advantage is uniformly broadcasted across the entire generated sequence. In contrast, DGPO’s reallocation mechanism dynamically focuses the optimization signal. We find that critical exploratory steps—such as pivotal mathematical substitutions or logical deductions—receive heavily amplified gradients. Conversely, standard grammatical tokens and redundant transitional syntax are naturally discounted. This visualization provides concrete evidence that DGPO interprets distribution deviation as a guiding signal for exploration, effectively resolving the severe temporal credit assignment problem inherent in critic-free RL frameworks.

5 Conclusion

In this paper, we introduce Distribution-Guided Policy Optimization (DGPO), a critic-free reinforcement learning framework that resolves the coarse-grained credit assignment and gradient instability issues of current RL paradigms. By replacing the unbounded Reverse KL divergence with a strictly bounded Hellinger distance, DGPO safely leverages distribution deviation as a guiding signal for exploration. We further propose a policy entropy gating mechanism to distinguish genuine reasoning leaps from hallucinatory noise by scaling this deviation against the model’s epistemic uncertainty. Extensive evaluations on the AIME 2024 and 2025 benchmarks confirm DGPO’s superiority. On the Qwen2.5-32B architecture, our fine-grained reallocation mechanism breaks the performance ceiling of standard outcome-based reward methods and substantially outperforms competitive baselines like DAPO. Ultimately, DGPO provides a highly stable, computationally efficient, and theoretically grounded approach to unlock the deep reasoning capabilities of large language models.

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A Theoretical Analysis and Proofs

In this section, we provide detailed mathematical derivations and theoretical proofs for the core mechanisms of the DGPO framework. Specifically, Appendix A.1 demonstrates that the gradient derived from the Hellinger distance remains strictly bounded under extreme exploration scenarios where the reference policy probability approaches zero, whereas the traditional Kullback-Leibler (KL) divergence invariably leads to gradient explosion. Appendix A.2 proves, from the perspective of stochastic approximation, that the policy entropy gating mechanism preserves the fundamental convergence properties of policy optimization.

A.1 Gradient Stability and Boundedness: Hellinger vs. Kullback-Leibler

As highlighted in Section 3.2, relying on the unbounded Reverse KL divergence penalty induces severe gradient instability during long-horizon generation tasks in RLHF. We provide a rigorous mathematical formalization of this phenomenon here.

Gradient Divergence of the Traditional KL Penalty. Assume at step t , the current policy is denoted as $\pi_\theta(a|s)$ and the reference policy as $\pi_{ref}(a|s)$. The gradient of the standard local objective function incorporating the Reverse KL penalty is given by:

$$\nabla_\theta \mathcal{L}_{KL} = \mathbb{E}_{a \sim \pi_\theta} \left[\nabla_\theta \log \pi_\theta(a|s) \cdot A_t - \beta \nabla_\theta \left(\log \frac{\pi_\theta(a|s)}{\pi_{ref}(a|s)} \right) \right] \quad (8)$$

Expanding the gradient of the penalty term yields:

$$\nabla_\theta KL(\pi_\theta || \pi_{ref}) = \sum_a \nabla_\theta \pi_\theta(a|s) \log \frac{\pi_\theta(a|s)}{\pi_{ref}(a|s)} + \sum_a \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s) \quad (9)$$

Since the gradient of a constant sum over the probability simplex is zero ($\sum_a \nabla_\theta \pi_\theta(a|s) = \nabla_\theta \sum_a \pi_\theta(a|s) = 0$), the equation simplifies to:

$$\nabla_\theta KL(\pi_\theta || \pi_{ref}) = \sum_a \nabla_\theta \pi_\theta(a|s) \left(\log \frac{\pi_\theta(a|s)}{\pi_{ref}(a|s)} + 1 \right) \quad (10)$$

When the model explores a novel token a^* during generation, if the prior probability assigned by the reference model to this token is infinitesimally small, i.e., $\pi_{ref}(a^*|s) \rightarrow 0^+$, the logarithmic term diverges: $\log \frac{\pi_\theta(a^*|s)}{\pi_{ref}(a^*|s)} \rightarrow +\infty$. Assuming the current policy network’s parameterization satisfies $\nabla_\theta \pi_\theta(a^*|s) \neq 0$, the gradient norm escalates infinitely: $\|\nabla_\theta KL\| \rightarrow \infty$. This demonstrates that even a minuscule parameter update towards a novel trajectory can cause optimization collapse due to unbounded gradient explosion.

Strict Boundedness of DGPO Gradients. In contrast, DGPO substitutes the volatile KL divergence with the bounded Hellinger distance $d_{i,t}$, utilizing it as a foundational guiding signal rather than a rigid penalty. The Hellinger distance is formally defined as:

$$d_{i,t} = 1 - \sum_{a \in \mathcal{V}} \sqrt{\pi_\theta(a|x, y_{<t}) \pi_{ref}(a|x, y_{<t})} \quad (11)$$

According to the Cauchy-Schwarz inequality, because both π_θ and π_{ref} are valid probability distributions, their Bhattacharyya coefficient satisfies $0 \leq \sum \sqrt{\pi_\theta \pi_{ref}} \leq 1$. Consequently, the distance metric is strictly bounded for any state-action space: $d_{i,t} \in [0, 1]$.

DGPO further mitigates noise by scaling this deviation via a policy entropy gating mechanism: $s_{i,t} = d_{i,t} \cdot \tilde{\mathcal{H}}_{i,t}^\kappa$. Given that the normalized Shannon entropy strictly lies within $\mathcal{H} \in [0, 1]$, the joint score is inherently bounded: $s_{i,t} \in [0, 1]$. Transforming $s_{i,t}$ via a sequence-length (T_i) scaled softmax yields the reallocation weight $w_{i,t}$:

$$w_{i,t} = T_i \cdot \frac{\exp(s_{i,t}/\tau)}{\sum_{j=1}^{T_i} \exp(s_{i,j}/\tau)} \quad (12)$$

By the algebraic properties of the softmax function, the weight is strictly constrained: $0 < w_{i,t} < T_i$. As established in Equation 5, the gradient of the DGPO objective can be approximated as:

$$\nabla_{\theta} \mathcal{L}^{DGPO} \approx \mathbb{E}[w_{i,t} \cdot A_i \cdot \nabla_{\theta} \log \pi_{\theta}] \quad (13)$$

We establish the upper bound of its gradient norm as follows:

$$\|\nabla_{\theta} \mathcal{L}^{DGPO}\| \leq \mathbb{E}[\|w_{i,t}\| \cdot |A_i| \cdot \|\nabla_{\theta} \log \pi_{\theta}\|] \quad (14)$$

Because the sequence-level advantage A_i is Z-score normalized (i.e., bounded variance or explicitly clipped), and the score function $\|\nabla_{\theta} \log \pi_{\theta}\|$ is bounded by the network’s Lipschitz constant under standard parameterizations, coupled with our proof that $\sup(w_{i,t}) \leq T_i$, the overall gradient norm is strictly capped by a finite upper bound $M < \infty$.

Even when $\pi_{ref}(a|s) \rightarrow 0$, the deviation metric $d_{i,t}$ smoothly saturates at 1 without singularity. The resultant gradient multiplier $w_{i,t}$ scales continuously, completely neutralizing the risk of gradient instability and alleviating mode-seeking conservatism.

A.2 Convergence Properties Preserved with Entropy Gating

In the DGPO framework, the coarse sequence-level advantage is dynamically reallocated to yield the local token-level advantage: $A_{i,t} = A_i \cdot w_{i,t}$. It is imperative to prove that introducing non-uniform, entropy-modulated weights $w_{i,t}$ does not disrupt the fundamental convergence guarantees of Actor-Critic or critic-free policy gradient methods.

Conservation of the Unbiased Advantage Expectation. Standard sequence-level methods (e.g., GRPO) uniformly assign a scalar advantage A_i across all tokens within a trajectory. In DGPO, the architectural design of the softmax reweighting strictly enforces a unit-mean property across the sequence:

$$\frac{1}{T_i} \sum_{t=1}^{T_i} w_{i,t} = 1 \quad (15)$$

This guarantees that the total mass of the reallocated credit across a given trajectory is perfectly conserved:

$$\sum_{t=1}^{T_i} A_{i,t} = \sum_{t=1}^{T_i} A_i \cdot w_{i,t} = A_i \sum_{t=1}^{T_i} w_{i,t} = A_i \cdot T_i \quad (16)$$

Consequently, under the trajectory expectation of the Markov Decision Process (MDP), DGPO maintains a positive inner product with the original unweighted policy gradient, ensuring the global update direction remains unbiased.

A.3 Adaptive Step-Size Perspective of Entropy Gating.

The policy entropy gating mechanism effectively acts as a state-dependent uncertainty regulator. According to the Robbins-Monro stochastic approximation conditions, stochastic gradient descent converges to a local optimum provided the effective learning rate (or scaling multiplier) remains non-negative and satisfies specific decay criteria.

In our formulation, the scaling weight is strictly positive ($w_{i,t} > 0$). Its behavior bifurcates based on epistemic uncertainty:

- **High-Entropy States (Deliberate Exploration):** When uncertainty is high ($\tilde{\mathcal{H}}_{i,t} \rightarrow 1$), $w_{i,t}$ is dominated by the valid distribution deviation $d_{i,t}$. The gating mechanism allows for amplified gradients, accelerating learning at these pivotal exploratory steps.
- **Low-Entropy States (Confident Hallucinations):** Conversely, if deviation is high but uncertainty is low ($\tilde{\mathcal{H}}_{i,t} \rightarrow 0$), the model is likely generating confident hallucinations. The entropy gate aggressively compresses the joint score $s_{i,t} \rightarrow 0$, resulting in a diminished $w_{i,t}$. Mathematically, this acts as an adaptive mechanism that decays the learning step size at spurious or overconfident tokens, preventing the reinforcement of false positive signals.

The entropy gating mechanism strictly modulates the relative update step sizes across the multi-dimensional trajectory space without altering the sign of the advantage function. It fully satisfies the convergence prerequisites of stochastic approximation. By systematically filtering out low-entropy, high-deviation noise (hallucinations), DGPO significantly reduces the variance of the gradient estimator, thereby achieving a smoother and theoretically robust optimization landscape.

B Limitations and Future Work

B.1 Limitations

While Distribution-Guided Policy Optimization (DGPO) provides a robust and theoretically grounded framework for fine-grained credit assignment, we acknowledge several limitations in our current study.

Domain Specificity. The empirical evaluations in this work predominantly focus on highly complex mathematical reasoning tasks, specifically utilizing the AIME 2024 and AIME 2025 benchmarks. While DGPO demonstrates state-of-the-art performance in eliciting mathematical Chain-of-Thought (CoT) generation, its efficacy in other critical alignment domains—such as open-ended creative generation, coding, and general instruction-following—remains to be extensively validated.

Hyperparameter Sensitivity. Our framework introduces two new hyperparameters: the reallocation temperature (τ) and the entropy gating scaling factor (κ). Although our sensitivity analysis demonstrates that DGPO is highly stable and achieves optimal credit assignment within moderate ranges (e.g., $\tau \in [0.5, 1.0]$), transitioning to entirely different tasks or model architectures may still require empirical tuning to prevent overly sparse updates or the accidental reinforcement of confident hallucinations.

B.2 Future Work

Building upon the foundations established by DGPO, we identify several promising directions for future research.

Broader Task Generalization. Future work should evaluate the DGPO framework across a wider array of reinforcement learning tasks for Large Language Models (LLMs), including algorithmic code generation, multi-turn conversational alignment, and complex agentic workflows where fine-grained credit assignment is equally critical.

Dynamic Hyperparameter Scheduling. Rather than relying on static values for the reallocation temperature (τ) and the entropy scaling factor (κ), future iterations could explore dynamic scheduling or adaptive, state-dependent mechanisms. Automatically decaying or increasing these parameters based on the global training step or the moving average of the sequence-level advantage could further stabilize training and eliminate the need for manual tuning.